

Unburied DAS on the Moon: Modeling Ground-to-Cable Strain Transfer

S. Probst¹, M. Serra Garcia² (?), J.O.A. Robertsson¹, C.M. Donahue³

¹Department of Earth and Planetary Sciences, ETH Zurich, Zurich, Switzerland

²AMOLF, Amsterdam, The Netherlands

³Earth and Environmental Sciences, Los Alamos National Laboratory, NM, USA

Abstract

Distributed Acoustic Sensing (DAS) measures dynamic strain along a fiber-optic cable, offering a robust, densely-sampled alternative to traditional sensors. To ensure good ground-to-cable coupling, cables are typically buried in a shallow trench. Unburied surface deployments are attractive for rapid-response terrestrial applications as well as extraterrestrial missions, such as on the Moon, where burial is impractical. However, unburied DAS often suffers from severely degraded signal quality, due to poor strain transfer from ground to cable. The physical mechanisms governing this transfer remain unquantified.

Here, we present the first analytical and numerical model of unburied DAS coupling, representing the draped cable as a series of suspended segments between discrete ground contact points. We demonstrate that strain transfer efficiency is fundamentally limited by bending stress relief, a process where suspended cable segments accommodate ground strain by bending rather than deforming axially. We introduce a novel dimensionless parameter Θ , governed by the ratio of the cable's initial gravity induced sag to its radius. We show that when a segment's sag exceeds a quarter of the cable's radius, ground displacement is absorbed by bending rather than being transferred as measurable axial strain. This framework directly predicts how mechanical properties, cable dimensions, pretension, and varying surface gravity affect coupling efficiency, providing quantitative guidelines for optimizing cable design and deployment strategies on both Earth and the Moon.

1 Introduction

Distributed Acoustic Sensing (DAS)

Distributed acoustic sensing (DAS) is a fiber-optic sensing technique that measures the dynamic axial strain along a cable by analyzing phase shifts in Rayleigh-backscattered laser light (e.g., Hartog, 2017; Lindsey & Martin, 2021). DAS records the elongation or

shortening of the optical fiber, leading to an inherently one-dimensional measurement of strain inline with the sensing cable.

Over the past decade, DAS has been adopted widely for borehole seismic applications such as vertical seismic profiling (Daley et al., 2013; Mateeva et al., 2014), and increasingly for surface deployments in which the cable is deployed in a shallow trench and coupled to the ground by burial and compaction (Abukrat et al., 2023; Hubbard et al., 2022).

For surface deployments, burial serves two functions: it shields the cable from wind-induced noise and ensures continuous mechanical contact with the ground, ensuring that the strain is transferred from the surrounding medium onto the cable.

Motivation for Unburied DAS: Lunar Application

However, burial is costly, time-consuming and requires excavation infrastructure that is unavailable or impractical in many environments. This makes unburied deployments a necessity for example in rapid-deployment scenarios such as recording aftershocks following a large seismic event.

Burial is also unfeasible in the challenging lunar environment, where the fine-grain and abrasive regolith poses large risks to infrastructure. DAS has been proposed for deployment on the Moon because its robustness and dense spatial sampling make it an attractive alternative to single-station seismometers. On the Moon, a rover could unroll kilometers of fiber-optic cable directly onto the surface, and record moonquakes, meteorite impacts, and active sources (Wu et al., 2024; Zhai et al., 2024). Understanding whether and how such a deployment can transfer ground strain to the fiber is therefore a problem of critical relevance.

Empirical Evidence of Poor Strain Transfer in Unburied Deployments

Empirical results widely show that unburied DAS deployments exhibit degraded signal quality relative to buried configurations. An et al. (2023) demonstrated this by comparing DAS recordings of traffic-induced vibrations on the same road using the same cable in two configurations, cemented to the surface and uncoupled, finding that the uncoupled cable recorded signals with significantly weaker amplitudes and lower signal-to-noise ratio (SNR). Similarly, Harmon et al. (2022) compared surface coupling strategies in a grassy field and found that an uncoupled cable could only capture hammer-source energy within approximately 10 m, while weighted configurations more than doubled this range. Zandanel et al. (2026) compared buried and unburied deployments in a lunar regolith simulant and found that unburied cables were able to record earthquakes but with lower amplitude compared to the buried configuration.

Part of the amplitude reduction in unburied deployments can be attributed to atmospheric noise. Wind exerts mechanical forces on the cable which contaminates the seismic signal (Hudson et al., 2021; Probst et al., 2026; Viens et al., 2025). However, poor signal quality persists even in wind-free environments: Probst et al. (2026) showed in controlled laboratory experiments that unburied cables recorded substantially lower amplitudes than buried cables even without any fan-induced wind. These observations indicate that poor ground coupling, and not atmospheric noise alone, is a fundamental challenge of unburied DAS.

Two-step Strain Transfer Framework

A cable resting on or buried in the ground does not transmit ground strain to the fiber

core perfectly. One component of this transfer, which we term cable-to-fiber transfer, concerns how strain is propagated through the multiple concentric layers of a cable’s cross-section (jacket, buffer, coating, cladding, and fiber core). This process is governed by shear-lag mechanics. Softer intermediate layers, such as the gel inside loose-tube telecommunications cables, attenuate the strain before it reaches the fiber core (Falcetelli et al., 2020; Hubbard et al., 2022; Reinsch et al., 2017; Tan et al., 2021). Empirical comparisons confirm this by consistently showing that tight-buffered cables yield higher strain transfer rates than loose-tube cables (Castongia et al., 2017; Forbriger et al., 2024). A second, less well-understood step concerns how the seismic strain field in the ground is transferred to the cable’s outer surface in the first place - what we call ground-to-cable transfer. For buried deployments, this step has been studied analytically and numerically. Zhang et al. (2020) developed a theoretical framework for the ground-backfill-cable interaction for distributed strain sensing in boreholes. Celli et al. (2023) studied cable-ground coupling for DAS using a spring model, finding that the stiffness of the material immediately surrounding the cable is the dominant factor, leading to amplitude amplification and phase delays.

Together, these two steps govern how the ground strain is first transmitted to the cable outer surface, and then propagated inward through the cable layers to the fiber. For buried DAS, the physical mechanisms of both steps are reasonably well characterized. This study concerns unburied cables, for which the first step in this chain - the ground-to-cable transfer - remains poorly understood.

With no overburden, the only force coupling the cable to the surface is its own weight. Contact is intermittent as the cable bridges over surface irregularities and touches the ground only at discrete points. The tangential force available to transmit axial seismic strain from the ground to the cable is limited by Coulomb friction at these contact points. Existing research on unburied DAS deployments is almost entirely empirical. Field experiments have shown that adding weights such as sandbags improves coupling (Harmon et al., 2022), though not consistently (Forbriger et al., 2024). Pressing a cable into snow improved coupling compared to draped deployment (Mjehovich et al., 2023).

Proposed Coupling Mechanisms and Open Questions

Probst et al. (2026) found that thicker and stiffer cables perform better in unburied configurations and proposed bending stress relief as a key mechanism: a cable spanning between contact points tends to bend rather than deform axially, so that the ground-induced axial strain is partially absorbed by cable bending rather than transmitted to the fiber. Thicker, stiffer cables resist this bending more effectively and could therefore record ground strain more reliably. However, this mechanism was proposed qualitatively and has not been quantified or linked analytically to cable parameters.

A second potential mechanism reducing the strain transfer in unburied deployments is frictional slip between the cable and the ground at the contact points. As noted by Celli et al. (2023), for accelerations well below 1 g, a DAS cable depends entirely on frictional contact for its coupling. If the amplitude of the ground-induced axial force exceeds the maximum static friction available at the contact points, the cable will slip and reduce the axial strain transmitted to the fiber. Both mechanisms - bending stress relief and frictional slip - are expected to depend on cable parameters including the bending stiffness EI , axial stiffness EA , mass per unit length $m = \rho A$, outer radius r , and the friction coefficient μ between the cable and the surface. Furthermore, both mechanisms are dependent on

gravity, which is much lower on the Moon. It is therefore crucial to understand how gravity impacts these mechanisms, to predict the coupling in a lunar environment. At present, no model exists that quantifies these dependencies and predicts how well a given cable couples to the ground in a surface deployment.

The goal of this paper is to address this gap by developing an analytical and numerical mechanical model for the strain transfer from the ground to the cable in an unburied DAS deployment. We derive solutions for the axial strain transferred from the ground to the cable, modeling uncoupled cable segments as beams suspended between two contact points. We introduce a dimensionless number - a ratio of the involved parameters - which directly predicts the strain transfer efficiency for a given deployment. Our results provide the first quantitative framework for understanding and optimizing unburied DAS coupling, with implications for cable design on Earth and on the Moon.

2 Analytical Strain Transfer Model

2.1 Treatment of Cable as Segments Suspended Between Ground Contact Points

An unburied, surface-draped cable is modeled as a sequence of suspended segments and contact points (Figure 1a). The spacing between the contact points between cable and ground depends on surface roughness and cable stiffness.

When strained, hanging segments might bend rather than deform axially. How the strain is transferred from the ground via the contact points to the cable itself is calculated using beam theory. The hanging cable segments are modeled as beams which are deformed by the movement of their endpoints, the contact points with the ground. The following derivations and the numerical implementation are studying a single suspended segment. The response of a DAS cable - averaging over the gauge length - is composed of the averaged response of many such segments with varying length.

We consider the cable segment as a doubly clamped horizontal beam with mechanical properties: undeformed length L_0 , radius r , cross-sectional area $A = \pi r^2$, moment of inertia $I = \pi r^4/4$, Young's modulus E , and density ρ .

Assumptions are made to allow an analytic treatment of the problem. A cable segment with no inherent bend is assumed, which is fully straight in the absence of external forces. The contact points of the segment are assumed to be level (i.e., on the same height). The solution is also limited to a purely inline strain field with wavelengths much larger than the cable segment. No slip at the contact points is assumed - the motion of the cable at the contact points is identical to the ground motion at these points. This also implies no interaction between adjacent suspended segments, since the behavior of each segment is fully dictated by the movement of its endpoints.

The hanging cable segment at rest is thus modeled as a beam with doubly-clamped endpoints subject to gravity (Figure 1b). Upon a displacement of the endpoints, the cable deforms to accommodate the applied stress. The deformation is limited to two modes for this simplified treatment. The validity of this is seen in the later comparison to the numerical, exact solution. The two deformation modes are axial elongation x_l and the first flexural bending mode x_f (Figure 1c and d). The transverse deflection is approximated as

$w(x) = x_f \phi(x)$, where the first symmetric mode for a doubly-clamped beam is chosen as

$$\phi(x) = \frac{1 - \cos\left(\frac{2\pi x}{L}\right)}{2}, \quad (1)$$

where L is the horizontal span between the endpoints. This satisfies the boundary conditions $w(0) = w(L) = 0$ (beam endpoints fixed), and $w'(0) = w'(L) = 0$ (beam endpoints orientation fixed). The maximum vertical deflection of the beam occurs at its midpoint, where $w(L/2) = x_f$.

DAS is sensitive to axial strain in the optical fiber and is therefore only sensitive to the part of the strain that is transferred to the cable as axial strain, and not the bending. The strain transfer efficiency describes the fraction of the axial strain which is transferred from ground to cable and is defined as $\frac{\delta x_l}{\delta L}$, where δx_l is the axial length change of the cable segment and δL is the change of the horizontal span.

2.2 Analytical Strain Transfer Efficiency

We calculate the axial strain transfer in the following steps: first, we solve for the initial shape under gravity alone by equating the gravitational forces and bending resisting forces. Second, we solve for an incremental deformation by linearizing the system about this sagged state and minimizing the energy to calculate how an imposed static strain is partitioned between bending and axial elongation modes. In the following derivation, we distinguish between the undeformed material arc-length L_0 and the horizontal span L only within the geometric constraint (Equation 4). For the evaluation of the coefficients C_1 and C_2 we adopt the small-strain approximation $L_0 \approx L$.

The arc-length of the beam is given by $S = \int_0^L \sqrt{1 + (w')^2} dx$. Assuming small slopes ($|w'| \ll 1$), expand to second order:

$$S \approx \int_0^L \left(1 + \frac{1}{2}(w')^2\right) dx = L + \frac{1}{2} \int_0^L (w')^2 dx. \quad (2)$$

Substitute ($w = x_f \phi$):

$$S = L + \frac{1}{2} C_1 x_f^2, \quad C_1 = \int_0^L (\phi')^2 dx. \quad (3)$$

The arc-length of the beam must equal the undeformed beam length plus axial elongation $S = L_0 + x_l$. Thus, bending reduces the effective horizontal span for a given axial length:

$$L = L_0 + x_l - \frac{1}{2} C_1 x_f^2. \quad (4)$$

For small axial strains the energy of the axial deformation U_l , and the bending energy U_f are

$$U_l = \frac{1}{2} \frac{EA}{L} x_l^2, \quad \& \quad U_f = \frac{1}{2} EIC_2 x_f^2, \quad C_2 = \int_0^L (\phi'')^2 dx. \quad (5)$$

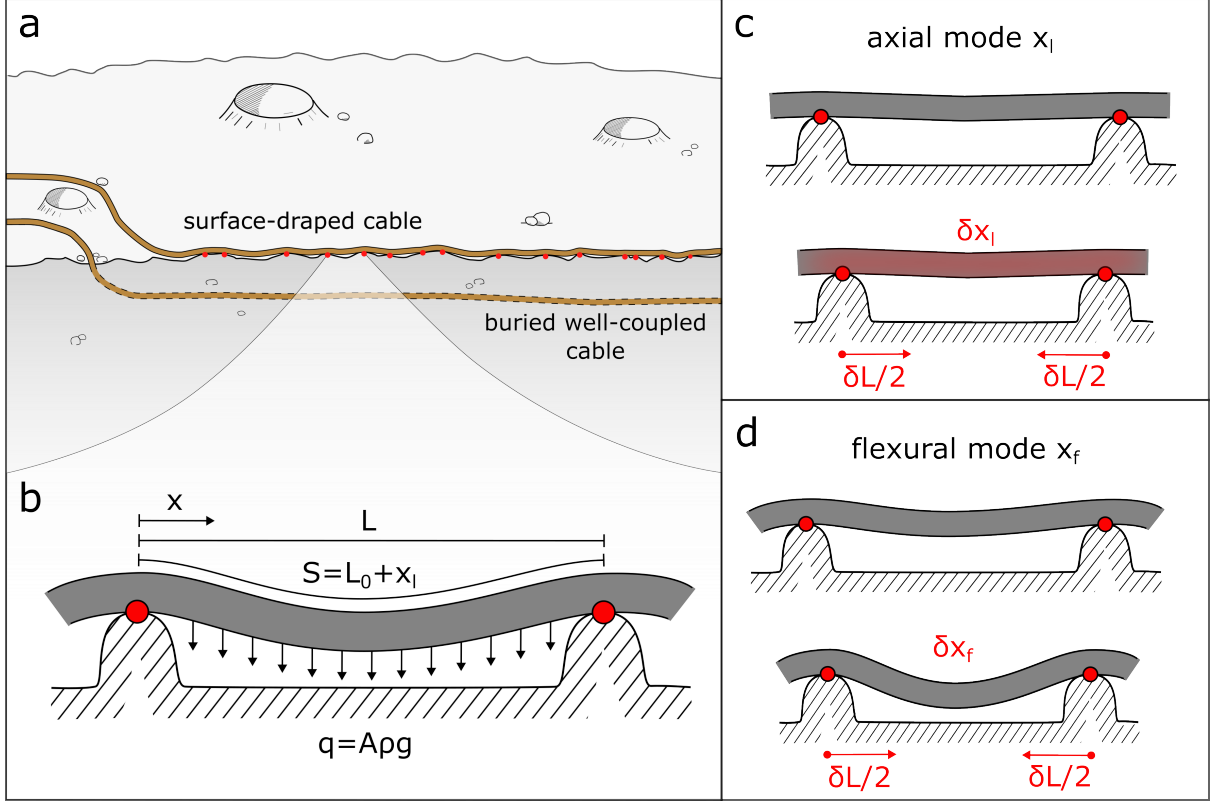


Figure 1: Unburied cables lack continuous ground contact, and form suspended segments between discrete contact points that accommodate applied ground strain through a combination of axial and flexural deformation modes. (a) Illustrative setup of a fiber-optic cable surface deployment in a lunar environment. (b) Zoom-in on a single suspended segment showing the variables used in the analytical derivations: the horizontal span between endpoints L , the arc-length S , the undeformed segment length L_0 , and the axial deformation x_l (change in arc-length). Gravity acts on the segment as a uniform distributed load $q = \rho A g$, leading to sagging. When the segment endpoints are deformed by an applied ground displacement δL , this strain is partitioned into two principal deformation modes. (c) The axial mode, where the cable accommodates the displacement by physically changing its arc length. The axial strain from the ground is transferred to the cable and can be measured by DAS. (d) The flexural mode, where the cable bends to accommodate the displacement without changing its actual arc length. The axial strain from the ground is not transferred to the cable and cannot be measured by DAS.

The total energy is then

$$U = U_l + U_f = \frac{1}{2} \frac{EA}{L} x_l^2 + \frac{1}{2} EIC_2 x_f^2. \quad (6)$$

The total potential energy of the cable under the load of gravity is $\Pi = U_f - W$, where W is the work done by gravity. This is under the assumption that the cable is under no initial axial tension, and therefore $U_l = 0$. Gravity is acting on the cable as a uniform distributed load ($q = \rho A g$) acting along the entire cable, therefore $F_g = \rho A g L$. Equilibrium occurs at

the minimum of the potential energy ($\frac{d\Pi}{dx_f} = 0$):

$$\frac{d}{dx_f} \left(\frac{1}{2} EIC_2 x_f^2 - \frac{1}{2} F_g x_f \right) = EIC_2 x_f - \frac{1}{2} F_g = 0. \quad (7)$$

This yields the equilibrium condition where the internal elastic restoring force matches the external gravitational force

$$EIC_2 x_{f0} = \frac{1}{2} F_g \Rightarrow x_{f0} = \frac{\rho AgL}{2EIC_2} = \frac{\rho AgL^4}{4\pi^4 EI}. \quad (8)$$

We consider a small ground movement δL , which causes small changes $x_f = x_{f0} + \delta x_f$ and $x_l = \delta x_l$. Linearizing about the initial state ($x_l = 0, x_f = x_{f0}$):

$$L = L_0 + x_l - \frac{1}{2} C_1 x_f^2 \Rightarrow \delta L = \delta x_l - C_1 x_{f0} \delta x_f \quad (9)$$

We minimize the incremental energy δU required to accommodate the ground movement δL :

$$\delta U = \frac{1}{2} \frac{EA}{L} (\delta x_l)^2 + \frac{1}{2} EIC_2 (\delta x_f)^2. \quad (10)$$

Substitute the expression for δx_l from the linearized constraint:

$$\delta U = \frac{1}{2} \frac{EA}{L} (\delta L + C_1 x_{f0} \delta x_f)^2 + \frac{1}{2} EIC_2 (\delta x_f)^2. \quad (11)$$

We minimize δU with respect to δx_f and solve for δx_f :

$$\delta x_f = - \frac{\frac{EA}{L} C_1 x_{f0} \delta L}{EIC_2 + \frac{EA}{L} (C_1 x_{f0})^2} \quad (12)$$

Then, plug δx_f back into the constraint:

$$\delta x_l = \delta L + C_1 x_{f0} \delta x_f = \delta L \frac{EIC_2}{EIC_2 + \frac{EA}{L} (C_1 x_{f0})^2} \quad (13)$$

This yields the axial strain transfer ratio,

$$\frac{\delta x_l}{\delta L} = \frac{1}{1 + \Theta}, \quad \Theta = \frac{EA}{L} \frac{(C_1 x_{f0})^2}{EIC_2} = \frac{\rho^2 g^2 A^3 L^8}{128 \pi^8 E^2 I^3}. \quad (14)$$

3 Numerical Methods

To numerically evaluate strain transfer, we model the suspended cable segment as a 1D Timoshenko beam. This theoretical formulation assumes linear elasticity and incorporates finite rotations and shear, allowing for the accurate simulation of large geometric deformations and coupling between transverse bending and axial elongation.

The beam model was implemented using the open-source finite element framework FEniCS. The numerical solution is obtained in two primary steps: initially, a nonlinear Newton solver is used to determine the static equilibrium shape of the beam under gravity, for two deployment scenarios where the endpoints are either clamped with initial tension or permitted to relax horizontally (as assumed in the analytical solution). Following this static step, we linearize around the gravity-sagged equilibrium position and analyze the dynamic behavior by solving for the natural vibration modes and performing a steady-state frequency response analysis with Rayleigh damping.

The mechanical properties and geometric parameters which govern the behavior of the Timoshenko beam as well as the tested ranges are detailed in Table 1. Every combination was modeled to comprehensively evaluate their collective influence on the resulting coupling mechanisms.

Table 1: [Final values used in simulations to be updated] The behavior of a cable segment modeled as a Timoshenko beam is governed by its mechanical properties (Young’s modulus E , shear modulus G , density ρ) as well as geometric parameters (radius r , segment length L) and gravitational acceleration g . Note that the shear modulus G is considered in the numerical model, since the model accounts for shear. However, for slender beams which cables are, the shearing is neglectable, and varying the shear modulus has very little impact on the results.

Parameter	Symbol	SI Unit	Used Unit	Min. value	Max. value	Increment
Cable radius	r	m	mm	0.5	10	0.5
Segment length	L	m	cm	0.5	10	0.5
Young’s modulus	E	Pa	GPa	$X*10^6$	$X*10^7$	$X*10^6$
Shear modulus	G	Pa	GPa	$X*10^6$	$X*10^7$	$X*10^6$
Density	ρ	kg m^{-3}	kg m^{-3}	X100	X1000	X100
Gravitational acceleration	g	m s^{-2}	m s^{-2}	1	10	1

4 Results

4.1 Initial Cable Sag Under Gravity

The initial sag of a suspended cable segment is governed by the balance between the downward pull of gravity and the cable's inherent bending stiffness. Assuming no initial tension, the analytical solution of the initial midpoint sag $w_0(L/2)$ (equal to the magnitude of the flexural mode x_{f0} derived in Equation 8) can be expressed as a ratio to the suspended length L to form a dimensionless sag parameter:

$$\frac{w_0(L/2)}{L} = \frac{x_{f0}}{L} = \frac{\rho Ag L^3}{4\pi^4 EI}. \quad (15)$$

The simplified analytical model accurately captures the fundamental equilibrium behavior: simulations agree closely with the linear analytical prediction at small to moderate sags Figure 2a. [Deviations at small Theta to be investigated. Is this a numerical imprecision or an actual phenomenon?] However, at large dimensionless ratios - representing highly compliant or long free-hanging segments - the numerical results diverge from the linear prediction because the small-sag approximation made in the analytical model breaks down. Furthermore, the numerical results demonstrate that an initial axial tension resists transverse deflection, restricting the initial sag to values lower than predicted by the tension-free analytical model.

4.2 Static Strain Transfer Efficiency

The efficiency with which static ground strain translates into axial cable strain is fundamentally controlled by Θ , a dimensionless parameter describing the geometrical and mechanical susceptibility of the segment to bending stress relief. By substituting the analytical solution for the initial sag $w_0(L/2)$ into the definition of Θ (Equation 14), this parameter simplifies to:

$$\Theta = \frac{1}{2} \left(\frac{w_0(L/2)}{r} \right)^2 \quad (16)$$

with the static analytical strain transfer efficiency:

$$\frac{\delta x_l}{\delta L} = \frac{1}{1 + \Theta} \quad (17)$$

These equations reveal that axial strain transfer is determined by how pronounced the initial sag is relative to the cable's radius.

The numerical simulations validate the analytical prediction that strain transfer efficiency decays rapidly for segments characterized by large Θ values (Figure 2b). When the initial sag is small compared to the cable's thickness ($w_0(L/2)/r \ll 1/4$ corresponding to $\Theta \ll 1/32$), the segment behaves like a straight rod, and the boundary displacement is transferred almost entirely as axial strain. Conversely, when the sag exceeds a fourth of the cable radius, the segment accommodates an endpoint displacement by simply increasing or decreasing the amount of bending, transferring almost no axial strain to the optical fiber. Notably, if the specific deployment configuration includes an initial axial pre-tension, the

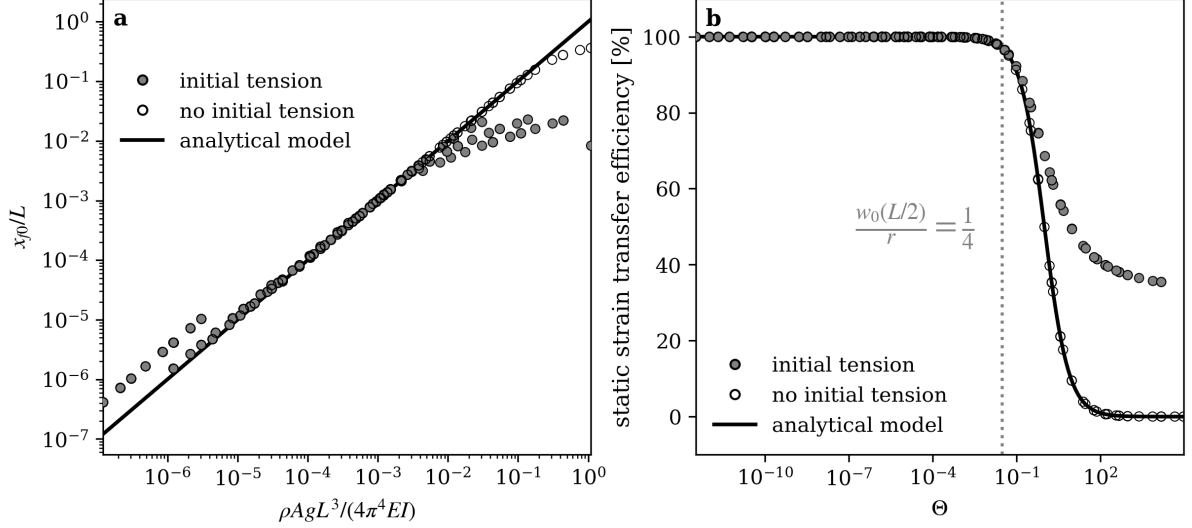


Figure 2: **Static behavior of an unburied cable segment under gravity.** The initial cable sag relative to the suspended segment length scales linearly with the ratio of gravitational to bending resisting forces, while the resulting static strain transfer efficiency is dictated by the dimensionless parameter Θ . **(a)** Initial dimensionless midpoint sag ($w_0(L/2)/L$) as a function of the force ratio. **(b)** Static strain transfer efficiency as a function of Θ , which depends on the ratio of the cable’s initial sag to its radius. Solid lines indicate analytical predictions from equations derived in Section 2.2, while markers represent numerical results. At small Θ , the cable behaves like a rigid straight rod, yielding near-perfect axial strain transfer. At large Θ , the cable behaves like a hanging string, transforming axial boundary displacements into changes in sag instead of transferring axial strain to the fiber. This efficiency loss at large Θ values is less severe for deployments with initial axial tension.

numerical simulations show that the strain transfer drop-off is less pronounced. Instead of decaying towards zero at large Θ regimes, the initial tension restricts the amount of initial sag, forcing the cable to maintain a higher axial strain transfer efficiency.

4.3 Dynamic Strain Transfer Efficiency

Natural Resonance Frequency

The dynamic response of a suspended cable segment is governed by its natural frequencies, most importantly the first flexural bending mode. This fundamental resonance frequency acts as a threshold dividing the cable’s behavior into quasi-static, resonant, and inertia-dominated regimes. For a straight, horizontal beam with rigidly clamped endpoints, the fundamental angular frequency is determined by its geometry and material properties:

$$\omega_1 = \frac{22.4}{L^2} \sqrt{\frac{EI}{\rho A}} \quad (18)$$

If the seismic frequency band of interest lies sufficiently far below this resonance, the cable segment will behave in a predictable, frequency-independent manner.

While the fundamental frequency is primarily governed by these inherent properties, the introduction of gravity causes a geometric stiffening effect that increases the resonance

frequency. A cable pre-bent by its own weight naturally exhibits a higher effective flexural stiffness than a perfectly straight counterpart. However, this gravity-induced frequency shift is generally of subordinate operational importance; the absolute magnitude of the shift remains small compared to the natural variations resulting from the primary geometric and mechanical properties of the cable.

Dynamic Cable Response

At seismic frequencies well below the fundamental resonance, the cable responds quasi-statically, yielding a dynamic strain transfer efficiency that matches the static limit (Figure 3). Because the driving ground motion is slow compared to the cable’s natural mechanical response time, internal inertial forces remain negligible. The segment is able to transition between equilibrium states in phase with the ground, maintaining exactly the same fractional partition between axial stretch and bending derived in subsection 4.2. As the driving frequency increases toward the fundamental resonance, the ground strain strongly couples into the transverse bending mode, manifesting as a peak in the midpoint displacement amplitude at the resonance frequency. Before reaching the resonance, the midpoint deflection is still close to in-phase with the ground movement. Consequently, ground displacement is absorbed into bending more effectively. This destructive coupling causes the strain transfer efficiency to drop below its static limit and reach a minimum just before the resonance frequency.

Traversing the resonance frequency the phase of the midpoint displacement changes to an almost 180-degree phase shift, fundamentally inverting how the bending mode and the axial mode interact. As the phase shifts past -90 degrees toward fully out-of-phase (-180 degrees), the cable starts moving in opposition to the ground driving motion: when the endpoints are pulled outward, the inertia of the cable causes it to bend further, rather than to straighten. This outward boundary pull combined with the bending of the cable constitutes a constructive coupling — the fiber is stretched both by the ground and by its own whipping motion. This amplifies the signal, causing the strain transfer efficiency to spike to well over 100% immediately above the resonance frequency. As seen in Figure 3, the severity of this amplification peak is highly dependent on Θ , with highly compliant, sagging cables experiencing massive efficiency spikes reaching multiple hundred percent efficiency.

At driving frequencies significantly exceeding the first flexural mode, structural inertia ultimately suppresses the bending mode. The transverse flexure becomes “too slow” to follow the high-frequency oscillations, causing the midpoint displacement to decay toward zero. While the much stiffer axial mode is still able to respond to the driving motion, the coupling between the modes becomes negligible. The cable segment is effectively forced into a purely axial deformation regime. Consequently, regardless of the initial baseline sag or stiffness defined by Θ , the dynamic strain transfer efficiency asymptotically approaches 100% in the high-frequency limit.

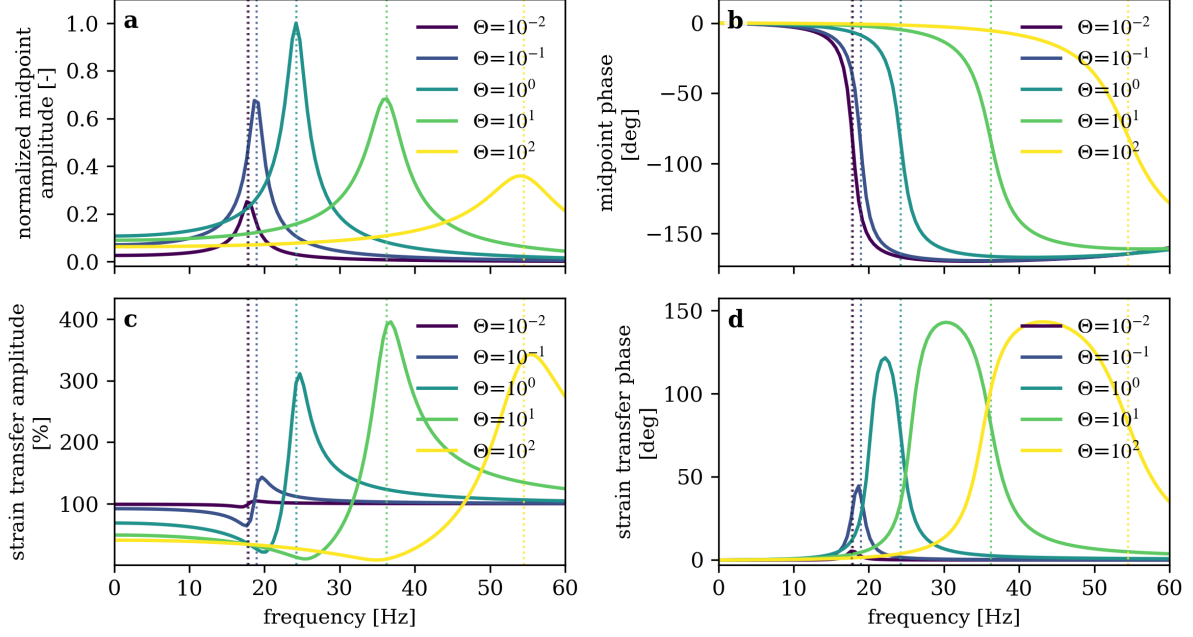


Figure 3: **Dynamic behavior of the axial and bending mode coupling and resulting strain transfer efficiency.** Midpoint deflection and strain transfer exhibit distinct resonance behaviors governed by the intensity of the flexural mode. **(a, b)** Amplitude and phase response of the cable’s midpoint deflection relative to its sagged state (not absolute deflection), measuring the intensity of the bending mode during axial deformation. The deflection shows a typical resonance peak and a frequency-dependent phase shift. **(c, d)** Amplitude and phase of the dynamic strain transfer efficiency as a function of frequency. At low frequencies, the strain transfer matches the static limit. As the frequency approaches resonance, efficiency is initially attenuated due to destructive mode coupling, followed by an amplification peak driven by constructive coupling as the phase rotates across the resonance. At high frequencies, cable inertia suppresses the bending mode entirely, resulting in near 100% strain transfer efficiency. The resonance peak systematically shifts to higher frequencies for larger initial sags (corresponding to higher Θ values): bending effectively stiffens the cable against the flexural mode, thereby raising its resonance frequency.

5 Discussion

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Discussion of the results

- The strain transfer efficiency of an unburied cable segment is entirely determined by the ratio of its gravity-induced sag to its radius, captured by the dimensionless parameter Θ together with the initial axial tension in the deployment. A cable segment with small sag relative to its cross-section is mechanically close to a straight rod. Without a sagging geometry to exploit, it can only accommodate axial endpoint strain through axial deformation. A cable with large sag relative to its radius absorbs a displacement of the endpoints more efficiently by changing the amount of bending, rather than by axial deformation.

- Deployments with $\Theta < 1/32$, corresponding to a sag of less than $r/4$, behave rod-like, while deployments with larger Θ show decreasing strain transfer efficiency.
- Deploying cables with initial tension mitigates some of the strain transfer loss. However, the cutoff Θ value is the same for deployments without and with tension, and the goal is to be below the cutoff in any case.
- In reality, the effect of the gauge length averages the response over many such segments - each mm to cm long - and produces a combined response. This means, that the output of each gauge length will vary depending on the average length of the segment within that gauge length.
- For dynamic strains approaching the resonance frequency of the suspended segment, the response becomes complicated involving large amplitude variations and phase shifts. This regime needs to be avoided to record reliable strains.

Limitations of the model

- Slip at contact points has not been considered in this model. This is an additional effect that might occur and reduce strain transfer.
- The analytical as well as numerical solutions assume cables are initially straight with no inherent bending. In practice, cables often have some bending, due to being stored on a spool. This has the potential to decrease the predicted strain transfer values, since it will effectively add to the gravity induced sag.
- The frequency response analysis assumes steady-state. However, in practice seismic signals are transient. Understanding the transient behavior is not crucial, since resonance behavior of the cables has to be avoided altogether, in order to measure meaningful results.
- Model assumes bare ground. Vegetation (e.g., grass, which lifts up the cables) will have additional transfer step and decrease strain transfer.
- Single-segment model

Implications for cable design and deployment

- To minimize Θ and maximize the strain transfer cables should have a large radius and be stiff, but light-weight. The ground surface should be as flat as possible and surface roughness minimized, to minimize the length of free-hanging segments.
- Strength members such as metal rods increase stiffness, however, they have been shown to create artifacts by acting as waveguides, and should therefore be avoided [source].
- Θ has extremely strong dependence on L , meaning that ensuring a deployment with as much ground contact as possible is paramount.
- Any initial bend of the cable promotes bending instead of axial deformation, and reduces strain transfer. It is crucial to use cables that don't maintain a bend from being on a spool.
- Lower gravity on the Moon reduces Θ and thereby the risk of poor strain transfer. However, it is important to note, that lower gravity reduces the normal force at the

contact points and makes frictional slip more likely.

6 Conclusion

[To be written]

Acknowledgments

[to be written]

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